

IEEE SMC • MIT BENGALURU • TECHNICAL SYMPOSIUM

The Pattern Machine

Your life and your city are running on loops you didn't write. Let's read the code — then ship the agent.

Every stable pattern is an empty slot where an agent can live. Your 2 a.m. scroll.

The surge price at 6 p.m. The bin you forget every Tuesday. This talk is about seeing those loops — and learning the one move that turns any of them into an agent.

Exploring Pattern Recognition in Speed AI Engineering the Agentic Future 5 SEP 2026 • AB5-308

40 slides • then you build one

A pattern you know intimately



Nobody:



Absolutely nobody:



My thumb, 2:47 AM: "last reel, then I sleep fr"

Funny because it's predictable. Same trigger, same routine, same reward, same 8 a.m. regret.

You already live inside pattern machines

7 hrs

Average daily screen time. Mostly autopilot, mostly unnoticed.

3 sec

Typical gap between "lock phone" and "unlock it again."

1.4×

Peak-hour surge multiplier your ride app quietly decided on.

None of this is a moral failure. It's **un-owned patterns** — and someone already built agents for them. The feed. The pricing engine. The notification scheduler.

Two kinds of people in an agentic world



PATTERN-TAKER

Lives inside agents other people built

The feed picks your evening. Autoplay picks the next one. Surge picks your budget. Your patterns are someone's KPI.



PATTERN-MAKER

Spots their own loops and builds agents for them

Reads "I always miss the recycling truck" as a spec. Wraps it in `perceive` → `decide` → `act`. Now the loop works *for* them.

This lecture is a 40-minute conversion from the left card to the right one.

ACT I

See the loop

Trigger → routine → reward. Spot it in your phone, your week, your street.

ACT II

The one move

Pattern + senses + a brain + hands + a loop = agent. Not model training.

ACT III

Gallery: your life

Five personal agents, built as blocks. Pick one to steal.

ACT IV

Gallery: your city

Surge, signals, crowds, grids – patterns at a million-person scale.

ACT V

Deterministic × not

The rails vs the judgment. Why every real agent is a hybrid.

— ACT I

A **pattern** is any regularity
stable enough to predict —
and act on before it happens.

TRIGGER → ROUTINE → REWARD. LEARN TO SEE THIS SHAPE EVERYWHERE.

Every app you love runs the same four beats

BEAT 1 Trigger Bored, anxious, lonely – or a buzz, a badge, a red dot.	BEAT 2 Action Smallest possible move. Unlock, swipe, tap. Zero friction.	BEAT 3 Variable reward Sometimes gold, sometimes nothing. The randomness is the hook.	BEAT 4 Investment A like, a streak, a follow – you load your own next trigger.
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Same four beats can point at a gym, a savings pot, a language app. Same machine – you just choose the target.



MONEY

The 11 p.m. cart



FOCUS

The 9-tab explosion



SLEEP

Revenge bedtime



SOCIAL

”I’ll reply properly later”

Willpower is not a pattern. It's the weather.

SUNDAY NIGHT: "NEW WEEK, NEW ME"



EXPECTATION: 6 AM, 5 days, meal-prepped, glowing



REALITY: Tuesday, raining, "rest week starts today"

Motivation arrives on a distribution — rarely at 6 a.m. on a cold Tuesday.

DON'T FEEL MORE MOTIVATED. BUILD THE THING THAT DOESN'T NEED YOU TO.

Some of your patterns are clockwork. Some are a dice roll.



DETERMINISTIC

Same every time

Alarm at 07:00. Coffee before words. Rent on the 1st. Predict with a rule.



NON-DETERMINISTIC

A distribution

Whether you train. What you eat when stressed. Your reply speed. Predict with a probability.

Great agents: **deterministic parts** for the rails, a **non-deterministic part**
for the judgment. Hold that.

Every self-improvement decision, ever



Rely on future-me having more discipline than present-me has ever had



Spend one afternoon building a system so future-me doesn't decide



Us, every Jan 1, sweating, pressing the first one

Agentic thinking in one line: move the decision out of the hard moment and into a calm afternoon.

— ACT II

**Pattern + senses + a brain
+ hands + a loop = agent.**

THAT'S IT. NO GPU. NO MODEL TRAINING. MOSTLY PLUMBING.

Four things you bolt onto a pattern



+ SENSES

Perceive

Read the signals you already generate.



+ A BRAIN

Decide

Given context, pick one useful move – or none.



+ HANDS

Act

A few small, well-labelled tools. Nothing else.



+ A LOOP

Learn

Did it work? Feed the outcome back.

The "brain" is one model call. The other three are code you could write today.

Perceive → decide → act → observe → repeat

01 Perceive Read state: sensors, calendar, the last result.	02 Decide Pick the next action. Rule, or model call.	03 Act Fire a tool. The world changed.	04 Observe Take in the result. Win? Error? New info?	05 Check Done – or loop with what you now know.
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A thermostat, a chess engine, a coding assistant, a traffic light: all run this. What differs is **how "decide" works** and **how tightly the loop is scripted**.

Why the “decide” step is a model, not an `if`

RULES WORK WHEN...

Every case was foreseen

`if days_since > 3: nudge()`. Fine for a week. Then you tune it out — a generic nag isn't a pattern-break, it's noise.

A MODEL WORKS WHEN...

The case wasn't foreseen

It's seen millions of traces of people planning, nudging, recovering. It produces a plausible next move for situations *nobody enumerated*.

- ▶ **Generalization** — handles the unseen Tuesday the way a classifier handles an unseen photo.
- ▶ **Language is the interface** — goals, context, tool calls are all just text it continues.
- ▶ **Steer with examples**, not a retrain. Change the prompt, change the behavior.

A loop with no stopping condition

AUTONOMOUS AGENT, ITERATION 47	
 "I'll just retry the same failing call one more time"	 "This is fine. Making progress. \$840 in tokens."
NO MAX-STEPS. NO BUDGET. NO HUMAN.	

The loop is the power and the danger. Unbounded iteration turns a small bug into a runaway bill.

- ▶ Hard step / time / token ceiling
- ▶ Loop & repeat detection
- ▶ Reversible actions, or ask a human
- ▶ A "give up and ask" branch

— ACT III

Same recipe. **Different pattern.** Five
agents you could ship this month.

BLOCKS, NOT ESSAYS. STEAL ONE.



AGENT 02

Sleep-On-It

Late-night cart over ₹2k → 12-hour hold before it lets you check out.



AGENT 03

Never-Forget

Contact's birthday in 3 days → drafts the message + a gift link.



AGENT 04

Reply-Draft

Text unread 18h from someone you care about → writes the reply, you hit send.



AGENT 05

Keep-Alive

The 2 a.m. scroll, as an agent

SENSES

screen-time by hour

SENSES

tomorrow's 1st alarm

SENSES

current app in use

BRAIN (MODEL)

"90 min past 22:00 + 6:30 alarm → pick the softest effective interruption"

HANDS

dim_lights()

HANDS

grayscale_screen()

HANDS

hide_feed(30min)

HANDS

play("rain")

RAIL

never fully locks the phone • always one-tap "not tonight"

Week 1 it guesses. Week 6 it knows grayscale + rain works on you and the lecture-y notification doesn't.

Day 40 with a life-agent

it's 23:40, you've got the 7am standup, and you've
rewatched the same 3 reels. screen's grey now. rain's on.

Wind-Down • 23:40

rude but ok

23:41

that's 5 nights this week under 15 min of doomscroll. your
feed misses you

23:41

It didn't add willpower. It removed the moment where willpower was needed.

SAME TRICK WORKS FOR SPENDING, REPLIES, CHORES, STUDY.

“You always forget Mom’s birthday” — solved in 20 lines

- ▶ **Sense:** contacts + a "close people" list you set once.
- ▶ **Brain:** given the person and your past messages to them, draft something that sounds like *you*, not a card.
- ▶ **Hands:** `draft_message()`, `suggest_gift(budget)`, `add_calendar_reminder()`.
- ▶ **Rail:** never sends automatically. It stages; you approve.

The one irreversible act — hitting send to a human — stays with you. Always.

```
# the whole agent, basically
for p in close_people:
    if days_until(p.birthday) == 3:
        draft = llm(
            "Write a warm birthday text in MY voice",
            history = messages_with(p),
            person = p,
        )
        gift = llm("3 gift ideas", person=p, budget=1500)
        # stage for approval — never auto-send
        inbox.stage(draft, gift)
```

—— CHECKPOINT • ACT III

Life admin, before and after

POV: IT'S 08:12 ON A TUESDAY

 the recycling truck pulls away.
your bin: still in the kitchen.
next pickup: in 14 days.
you: "I'll remember next time" (you will not)

THIS EXACT SCENE HAS 2 BILLION RERUNS WORLDWIDE

A Bin Agent is embarrassingly simple: a council calendar feed, a Sunday-night nudge, one rail – "max one reminder."

BORING AGENTS ARE THE ONES THAT ACTUALLY CHANGE YOUR LIFE.

The pattern behind all five

1

One sensor you already have

Screen-time export, a calendar feed, a manual check-in.

2

One tool

Usually just "send me a message." That's a complete agent.

3

One rail

"Once a day." "Never auto-send." "Nothing over ₹500."

You don't need five. You need **one, tonight.**

— ACT IV

Zoom out. The **same loop** runs a city of ten million people — and the agents are already deployed.

YOU RIDE THEM EVERY DAY. NOW YOU'LL SEE THEM.

The price that reads the crowd

PATTERN

demand spikes at predictable times & places (rain, concert-out, 6pm)

SENSES

open app sessions

SENSES

nearby driver count

SENSES

weather • events • time

DECIDE (NON-DETERMINISTIC)

set a multiplier that pulls just enough drivers in without killing demand

RAILS (DETERMINISTIC)

hard cap on multiplier • frozen during declared emergencies • shown before you book

A perceive → decide → act loop, running per-neighbourhood, every few seconds. You've been inside this agent's decision hundreds of times.

The agent, from the back seat



"it started drizzling"



the app, instantly: "2.3× — surge
in your area"

IT DIDN'T PANIC. IT RAN THE LOOP.

You feel personally attacked. It's just a pricing agent reading a demand pattern and pulling a lever.

Green lights that watch the queue

- ▶ **Pattern:** traffic flow has strong daily & weekly rhythm, plus shocks (a match, a crash).
- ▶ **Senses:** inductive loops in the road, cameras, connected-vehicle pings.
- ▶ **Decide:** extend this green, hold that red, offset the next junction so a "green wave" forms.
- ▶ **Rails:** minimum green time, pedestrian phases guaranteed, emergency-vehicle pre-emption overrides everything.

WHY IT'S AN AGENT, NOT A TIMER

It observes, then adapts

A fixed-timer light is deterministic and dumb. An adaptive corridor perceives the real queue and re-plans every cycle – a loop, at a junction, all day.

Cities running these report meaningfully shorter trip times on managed corridors – the loop, paying rent.

Moving a stadium without a stampede



SENSE

Density

Phone-network heatmaps, gate counts, platform CCTV occupancy.



DECIDE

Meter & reroute

Hold this gate, open that exit, delay a train 90s to spread the surge.

RAIL



Safety floors

Density thresholds that trigger humans, not just the model. Never optimize past them.

Same five parts as your Wind-Down agent. Just... 60,000 people instead of one sleepy student.

The loop that keeps the lights on

PATTERN

demand peaks are forecastable; supply from solar/wind is not

SENSE

live load

SENSE

weather → renewables

SENSE

price • reserves

DECIDE

pre-cool buildings, shift EV charging, page flexible factories — before the peak

RAILS

hospitals & critical loads never curtailed • frequency limits are hard law

Deterministic where being wrong means a blackout. Non-deterministic where it means a slightly warmer office.

Seeing your city



"the light turned green because it's my turn"



"the light is on a timer"



"the light sensed my lane's queue and re-planned the corridor"



"I live inside a few thousand perceive→decide→act loops and now I can name them"

Same idea at every scale: spot the regularity, act on it before it acts on you.

THE PERCEPTRON DID IT FOR PIXELS. THE GRID DOES IT FOR GIGAWATTS.

— ACT V

Deterministic: same
input, same output.

Non-deterministic: same input, a
distribution.

NEITHER IS BETTER. THEY FAIL DIFFERENTLY. THEY BELONG IN DIFFERENT PLACES.

Deterministic
Non-deterministic
BEHAVIOR
Fixed path. Reproducible bit-for-bit.
Sampled from a probability distribution.
BUILT FROM
Rules, lookups, typed functions.
A learned model; search with randomness.
SUPERPOWER
Auditable. Testable. Zero-error tolerance.
Handles the unforeseen & the ambiguous.
FAILURE MODE
Silent gap – the case no one wrote a rule for.
Confident wrongness – plausible but false.
TEST WITH

Money movement



Access & permissions



The agent's budget & kill-switch



Anything an audit will replay



Tool-argument & schema checks



Understanding a free-text ask



Planning steps for a novel task



Drafting — text, code, replies



Choosing which tool fits



Judging relevance & ranking



Every real agent: soft core, hard shell

HARD SHELL • IN

validate • authenticate • route • rate-limit

SOFT CORE

model reasons, plans, drafts, picks a tool

HARD SHELL • EVERY ACTION

type-check args • permission-check • dry-run the reversible ones

HARD SHELL • OUT

verify vs criteria • log • gate • (human sign-off on the irreversible)

The model *proposes*. Plain, tested code *disposes*. It can't hallucinate its way past the shell.

Putting the layer where it belongs

"SHOULD THE AGENT DECIDE THE PAYMENT AMOUNT?"	
 let the vibes-based language model freestyle a transfer, temp 0.8	
 a 3-line function: capped, once-a-day, cancellable, logged	
FLIP IT FOR "WRITE SOMETHING THAT GETS ME OFF THE COUCH"	

Maturity isn't "we use agents everywhere." It's being able to point at any component and say why it's deterministic or not.



VISIBLE

Show the "why now"

Can't watch it think → can't trust it, and you'd be right.



BOUNDED

Hard rails on anything costly

Caps & allow-lists in code, not prompt text.



REVERSIBLE

Every action has an undo

Mute, snooze, cancel – in one tap.



CONSENSUAL

You're the principal

Two ways to leave this room



"I'll just try harder next week"
— every week since 2019



"I mapped one loop and wired a
30-line agent to it tonight"

BE THE SECOND ONE. GENUINELY ONE EVENING.

No lab, no GPU, no research budget. One honest look at a pattern, an API key, a cron job.

LOW BAR. COMPOUNDING PAYOFF.

Ship the smallest possible life-agent

The brief

- ▶ **Pick the pattern that stung** in slide 8. One only.
- ▶ Write its loop in one line: `trigger → routine → reward`.
- ▶ **One sensor.** Anything real — an export, a calendar, a check-in.
- ▶ **One tool.** "Send me a message" counts.
- ▶ **One deterministic rail.** "Max once a day."
- ▶ Run `perceive → decide → act` once by hand. Then cron it.

Three sentences to leave with

- ▶ Every stable pattern is an empty slot where an agent can live.
- ▶ An agent is a pattern in a loop, with senses and hands.
- ▶ Soft core, hard shell — the model proposes, plain code disposes.

Don't automate your whole life. Automate *one rainy Tuesday*.

— ONE LINE TO TAKE HOME

Stop being the pattern. Start building the agent.

Seventy years ago a machine learned to notice a shape it had never seen. Tonight you can build one that notices you're about to skip the gym, miss the bin, or overspend at midnight — and quietly moves the pieces so you don't.

Same lineage. Much shorter commute.

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